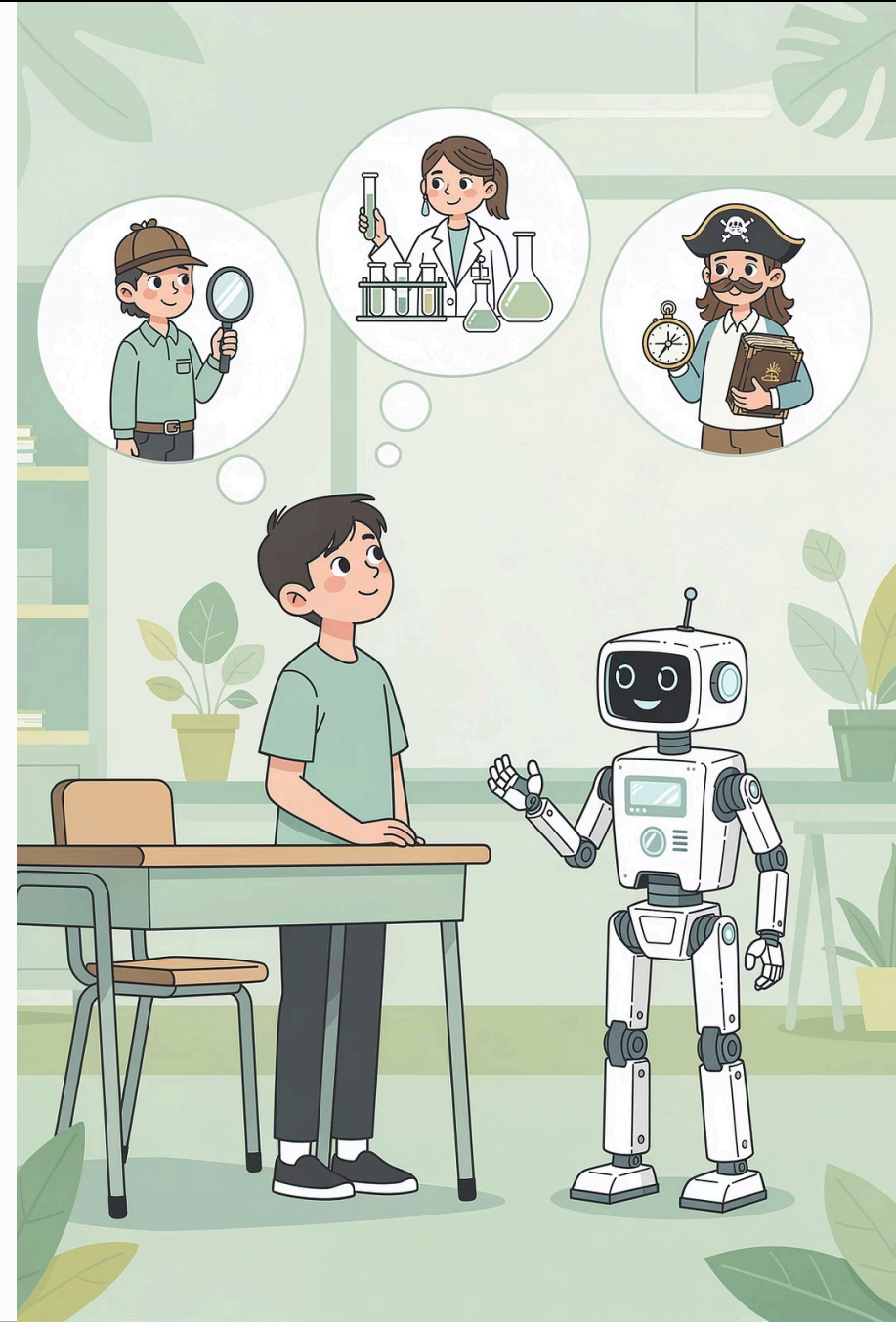


Session 12 : Advanced Prompting Techniques

You know how to give clear instructions, add context, and avoid common mistakes. Now it's time to take your skills to the next level. What if you could make AI act like a detective, a scientist, a pirate, or even a time-traveling historian? What if you could teach it with examples and ask it to show its reasoning step by step?

Today, you'll discover advanced techniques that give you even more control over AI outputs. Get ready!



Objectives

01

Zero-shot Prompting

Ask AI to do something without giving any examples.

02

Few-shot Prompting

Give one or more examples to teach AI the format you want (including one-shot).

03

Role-based (Persona) Prompting

Make AI act as a specific character.

04

Chain-of-thought Prompting

Ask AI to explain its reasoning step by step.

05

Combine Approaches

Mix multiple techniques for complex tasks.

06

Compare Techniques

Use all four on one task and decide which works best.

07

Add to Your Library

Save your best advanced prompts in My Prompt Library.

Recap: What We Learned in Session 11

Key Term	What It Means
Prompt	The instruction you give to an AI.
Instruction	A clear direction that tells the AI what action to take.
Context	Background information that helps the AI give a better answer.
Constraint	A limit or rule you give to AI (e.g., length, tone, style).
The Prompt Framework	[Action] + [Topic] + [Context] + [Format] + [Constraints]

- ① There's no single "perfect" way to write a prompt. Use helpful elements (instruction, context, format, constraints) when they make sense. Always iterate and stay in control. Today, we'll add powerful techniques to your prompting toolkit.

Warm-Up

Think about the last time you had to explain something to a friend - maybe how to play a new game, how to find your house, or how to make your favorite snack. What made your instructions clear? Did you just say "go to my house," or did you give landmarks? Did you just say "mix ingredients," or did you show them step by step?

When you give instructions, you naturally use different approaches:

Show Exactly What You Want

Like a picture of the final snack.

Pretend to Be Someone Else

"Imagine you're a chef..."

Walk Through the Steps

"First, do this, then that..."

Today we'll learn how to use these same ideas to give AI better instructions. These are the **advanced prompting techniques!**

The Advanced Techniques at a Glance

Just like explaining something to a friend, AI responds better when you use the right approach. Today's four techniques map directly to how humans naturally communicate instructions.



Zero-shot

Just ask no examples needed.



Few-shot

Show examples to set the pattern.



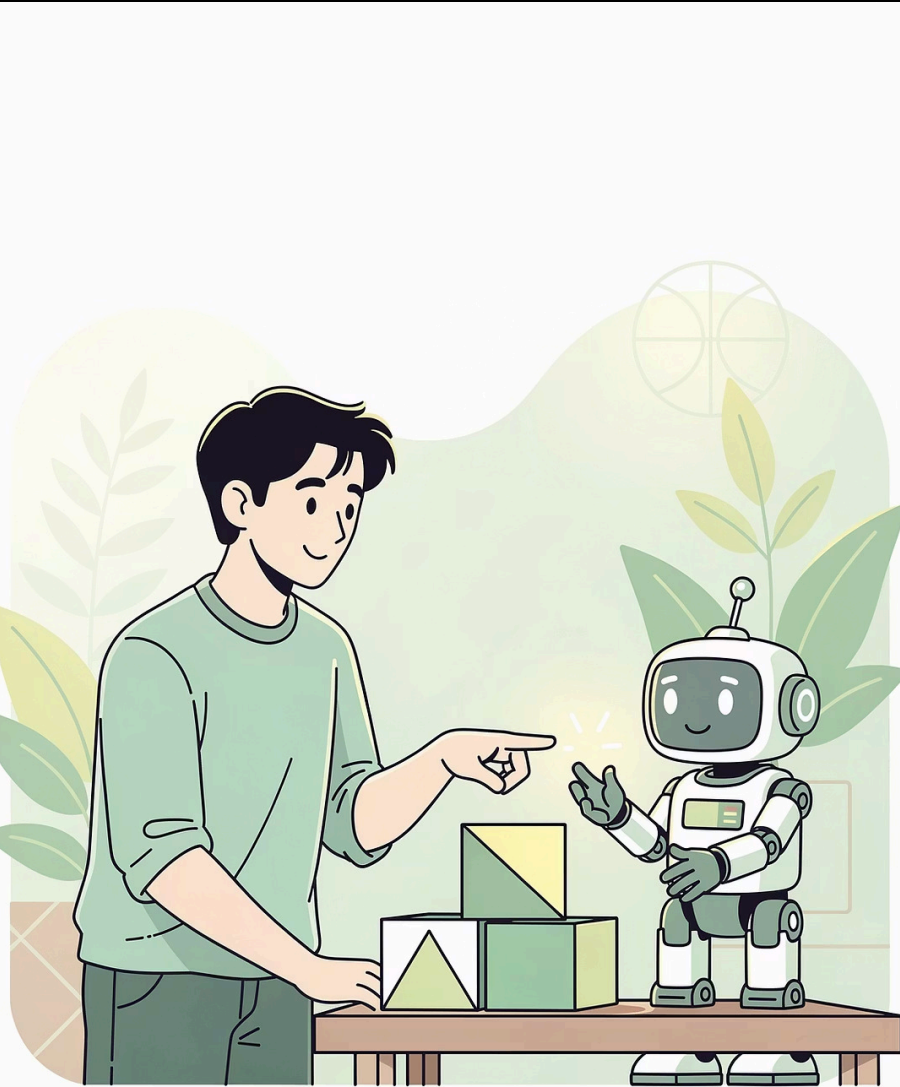
Role-based

Give AI a character to play.



Chain-of-thought

Ask AI to show its reasoning.



Zero-shot Prompting

Zero-shot prompting means giving the AI a task with **no examples** - just a clear instruction. You trust the AI to figure out the format and approach on its own.

What It Is

A direct instruction with no prior examples provided.

When to Use It

For simple, well-known tasks where the AI already has strong knowledge.

Example

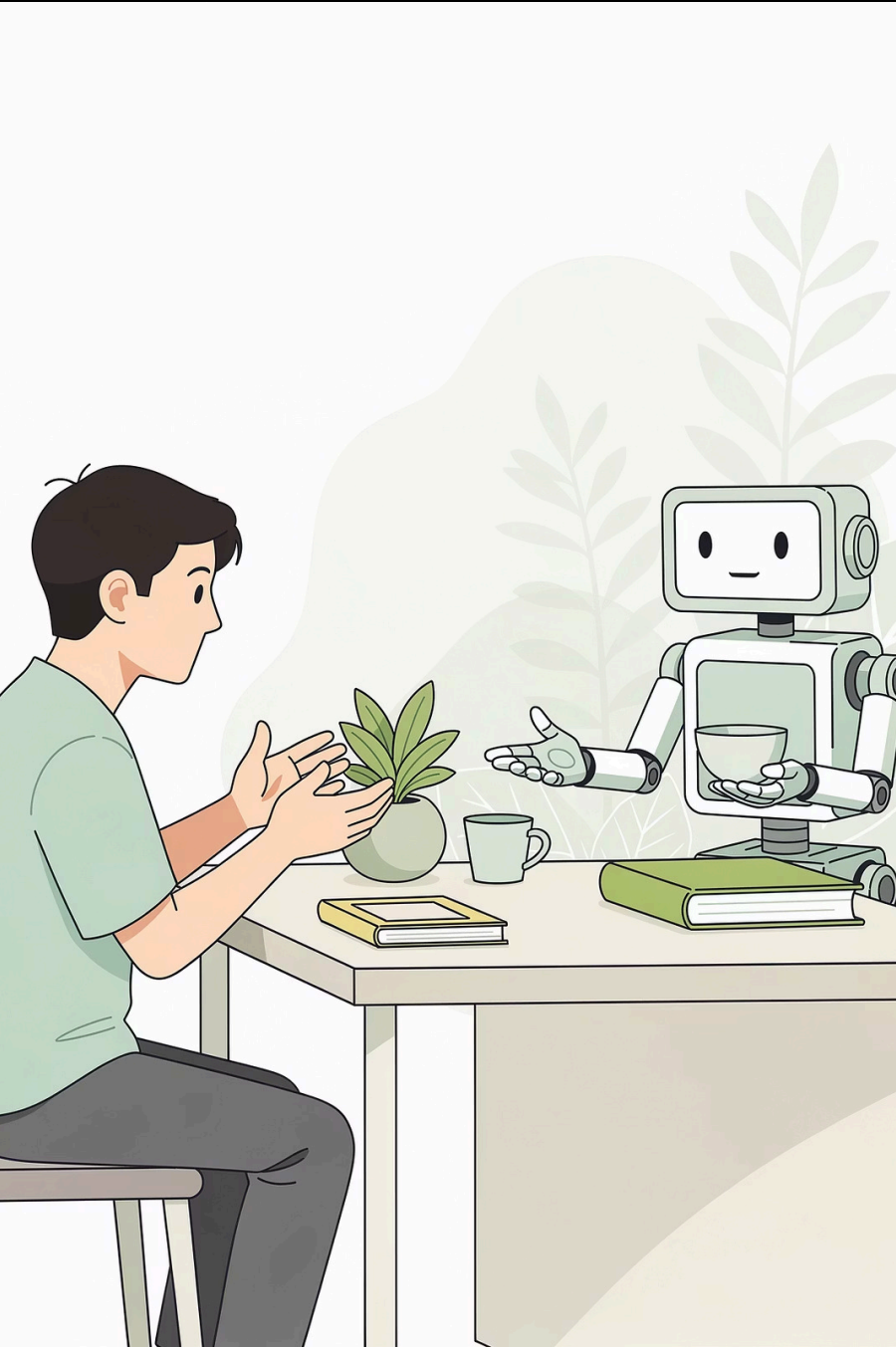
"Translate the following sentence into French: 'The sky is blue.'"

Zero-shot Prompting - Try It Yourself!

Your Turn: Write a zero-shot prompt for each task below. Remember: no examples, just clear instructions.

Task	Your Zero-shot Prompt
Ask AI to write a comic about a thunderstorm.	
Ask AI to give you three fun facts about Saturn.	
Ask AI to explain how to fold a paper aeroplane in two sentences.	

 **After writing, try them with Gemini or ChatGPT.** Did the AI understand what you wanted? If not, think about what details you could add – that's where few-shot prompting comes in next!



Few-shot Prompting

Few-shot prompting is when you give the AI **one or more examples** of what you want *before* asking it to do a similar task. The AI learns the pattern from your examples.

One-shot

One example given for the task.

Few-shot

Two or more examples given for the task.

Why it works: AI is very good at recognising patterns. When you show it the format, style, or structure you like, it will reproduce it.

Examples

One-shot Example

"Here's an example: For the word 'sun', I want a phrase: 'warm sun'. Now do the same for the word 'moon'."

Few-shot Example

"Here are some examples: 'cloud' → 'cloud dreams', 'river' → 'river flows', 'star' → 'star shines'. Now create similar phrases for the word 'forest'."

When to use: When you want the output to follow a specific format, style, or structure that might be unusual or not obvious from a zero-shot prompt.

Few-shot Prompting - Try It Yourself!

Your Turn: Write a few-shot prompt for each task. Include examples that show the exact style you want.

Task	Your Few-shot Prompt
Create a playful nickname for a pet hamster. (Give 2 examples first.)	
Write a short, dramatic movie title for a story about a lost key. (Give 1 example first.)	
Describe a forest using only three words that evoke mystery. (Give 2 examples for other places first.)	

-  **After writing, test them.** Compare the output with what you got from zero-shot. Do the examples help the AI match your style?

Mini Activity 1: Zero-shot vs Few-shot

Instructions: For each prompt below, identify whether it uses **Zero-shot** (no examples) or **Few-shot** (one or more examples). If it's few-shot, note how many examples.

Prompt	Technique (Zero / Few)	If few, how many examples?
"Write a short story about a squirrel who loves to dance."		
"Here's an example: 'sun' → 'sunny day'. Now do the same for the word 'rain'."		
"Give me three ideas for a weekend craft project."		
"Example: 'cat' → 'playful cat', 'dog' → 'loyal dog'. Now do the same for 'hamster'."		

Mini Activity 1: Answer Key

Prompt	Technique	If few, how many?
Write a short story about a squirrel who loves to dance.	Zero-shot	–
Example: 'sun' → 'sunny day' (one example)	Few-shot (one-shot)	1
Give me three ideas for a weekend craft project.	Zero-shot	–
Example: 'cat' → 'playful cat', 'dog' → 'loyal dog' (two examples)	Few-shot	2

Role-based (Persona) Prompting

Role-based prompting (also called persona prompting) asks the AI to adopt a specific character or identity - a teacher, a detective, a video game character, a famous scientist, a talking animal, etc.

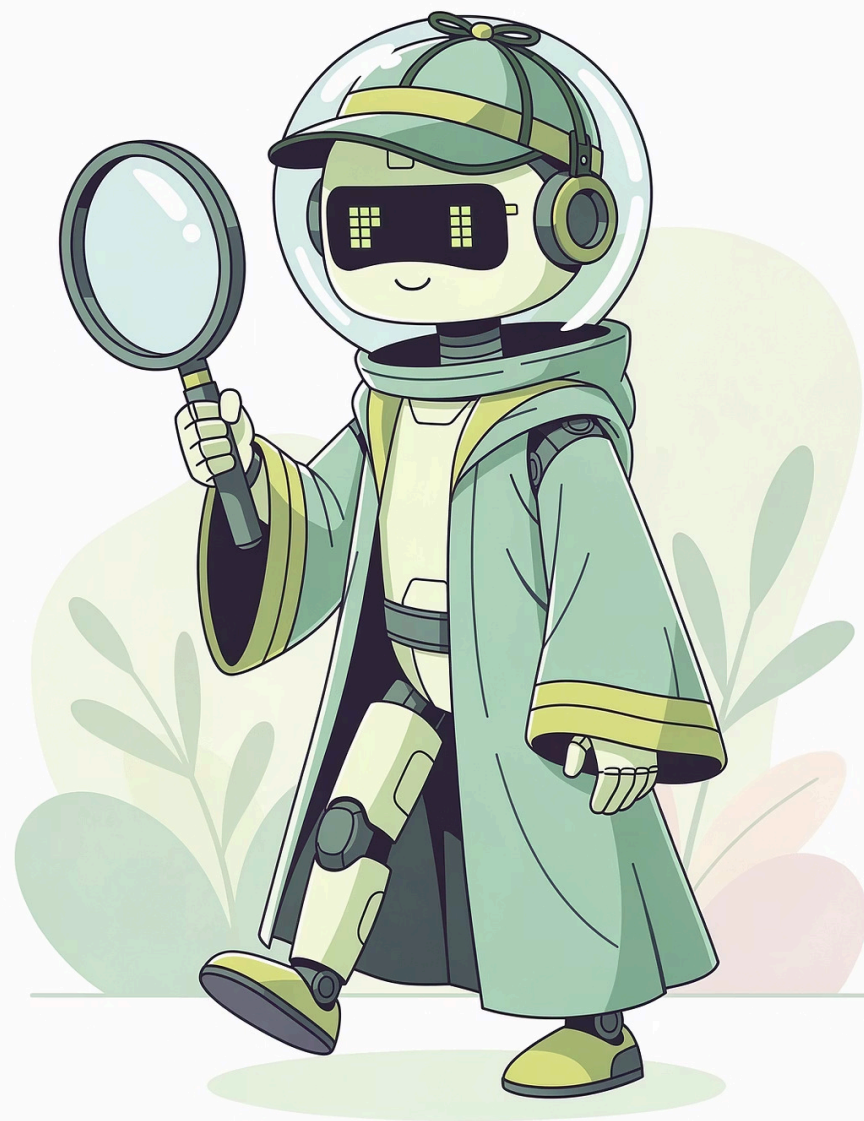
Why it works: The AI has been trained on millions of conversations in many different voices. When you give it a persona, it shifts its language, tone, and knowledge to match that role.

When to use

When you want a more engaging, creative, or tailored explanation. Role-based prompts are also great for exploring different perspectives.

Key insight

The persona gives the AI a "voice" - it makes the output more fun and sometimes more memorable.



Role-based Prompting: Examples

Without Role	With Role
"Explain how a rainbow forms."	"You are a friendly weather wizard who loves colors. Explain how a rainbow forms using a magical story."
"Give me tips for keeping a journal."	"You are a time-traveling librarian from the year 2150. Give me tips for keeping a journal, using futuristic examples."
"Write a poem about the ocean."	"Act as a mermaid who has lived under the sea for a hundred years. Write a poem about the ocean from your underwater perspective."

Mini Activity 2: Create Persona Prompts

Instructions: For each task below, write a **role-based prompt** that asks the AI to act as a specific persona. Be creative!

Task	Your Persona Prompt
Explain how a bird builds a nest.	<i>"You are a wise old owl..."</i>
Give advice on how to stay calm before a big test.	<i>"You are a friendly yoga instructor..."</i>
Describe the stars in the night sky.	<i>"You are an astronaut floating in space..."</i>

Mini Activity 2: Sample Answers

- **Bird builds a nest**

"You are a wise old owl who has watched many birds build nests. Explain how a bird builds a nest, using gentle, wise language and a few chirps of encouragement."

- **Staying calm before a test**

"You are a friendly yoga instructor who loves helping kids feel calm. Give three tips for staying calm before a big test, using slow, relaxing words."

- **Stars in the night sky**

"You are an astronaut floating in space, looking out your window. Describe what you see when you look at the stars, using wonder and excitement."

Chain-of-thought Prompting

Chain-of-thought prompting asks the AI to **explain its reasoning step by step** before giving the final answer.

How to do it

Add phrases like:

- *"Think step by step."*
- *"Explain your reasoning."*
- *"Show your work before giving the final answer."*
- *"Let's solve this together, one step at a time."*

Why it works

- It helps the AI arrive at a more accurate answer because it breaks the problem into smaller steps.
- You can check the reasoning for errors.
- You learn *how* the AI arrived at the conclusion which helps you understand the topic better.



Chain-of-thought: Prompt Examples

Examples - What YOU type (the prompt):

Without Chain-of-thought (prompt)	With Chain-of-thought (prompt)
"There are 12 cookies on a plate. You eat 4, and then your friend gives you 3 more. How many cookies are there now?"	"There are 12 cookies on a plate. You eat 4, and then your friend gives you 3 more. Think step by step and show your work. How many cookies are there now?"
"Should I bring an umbrella today?"	"The weather forecast says 70% chance of rain in the afternoon. I will be outside from 1 PM to 4 PM. Think step by step. Should I bring an umbrella today?"

 **Notice:** The chain-of-thought version adds a phrase like "Think step by step and show your work" to the original question. The AI then produces a response that includes the reasoning steps.

Chain-of-thought Prompting - Try It Yourself!

Your Turn: Convert each simple prompt into a chain-of-thought prompt by adding a phrase like "Think step by step" or "Explain your reasoning."

Simple Prompt	Your Chain-of-thought Prompt
"There are 15 birds on a fence. 6 fly away. Then 4 more land. How many birds are on the fence now?"	
"Is it better to do homework first or play video games first?" (Add a short scenario: you have 2 hours before dinner.)	
"If you flip a coin three times, what's the chance of getting heads every time?"	

📄 **Try them with Gemini or ChatGPT.** Compare the answers with and without chain-of-thought. Which one helps you understand *why* the answer is correct?



Combining Approaches

Combining approaches means using two or more advanced techniques in one prompt - for example, a role-based prompt that also uses few-shot examples and asks for chain-of-thought reasoning.

Why it works: Different techniques bring different strengths. Combining them can give you a powerful, tailored response that is both creative and accurate.

📌 **Key insight:** Combining techniques is like using multiple tools in a toolbox. The right combination can make your prompt much more powerful.

Combining Approaches: Examples

Role-based + Few-shot

"You are a friendly park ranger (role). Here's an example of a nature fact: 'The giant sequoia tree can live over 3,000 years.' Now give me three more interesting nature facts in the same style."

Chain-of-thought + Role-based

*"You are a detective trying to figure out who ate the last cookie. **Explain step by step** how you would investigate, using clues from the kitchen."*

Few-shot + Chain-of-thought

"Here's an example of solving a simple problem: 'If you have 5 stickers and you give 2 away, how many left? Step 1: Start with 5. Step 2: Subtract 2. Step 3: $5 - 2 = 3$. Answer: 3.' Now solve this problem step by step: 'A box has 12 pencils. You add 8 more. Then you take out 5. How many pencils are left?'"

All Three Combined

*"You are a nature guide (role). Here's how you explain a plant fact: 'A sunflower follows the sun from east to west during the day.' (example). Now explain how a cactus stores water, **step by step** (chain-of-thought)."*

When to use: When a single technique isn't enough, or when you want the best of multiple worlds - creativity, structure, and reasoning.

Hands-on Activity: Advanced Prompt Playground

Goal: Apply all four advanced techniques to **one task**, compare the outputs, and decide which technique (or combination) works best and **why**.

Setup: Use Gemini (or ChatGPT). Open your **My Prompt Library** document - you'll add your best prompts from this activity.



Choose Your Task

Task: Pick **one** of the following tasks:

1

Plan a Picnic

Plan a picnic in a park (e.g., what to bring, where to sit, games to play).

2

Mythical Creature

Create a description for a new mythical creature (e.g., a friendly creature that lives in a library).

3

Logic Puzzle

Solve a logic puzzle (e.g., "A dog, a cat, and a bird are in a house. The dog is not on the rug. The cat is on the couch. Where is the bird if it is not with the dog?")

4

Curious Fact

Explain a curious fact (e.g., "Why do we yawn?")

Try Each Technique

Now, try each technique with your chosen task:



Zero-shot

Write a simple prompt with no examples, no persona, no reasoning instruction.



Few-shot

Give 1-2 examples of the kind of output you want, then ask for a new one.



Role-based

Add a persona (e.g., "You are a friendly outdoor expert," "You are a mythical creature expert," etc.).



Chain-of-thought

Add a phrase like "Think step by step" or "Explain your reasoning" to your prompt.

Optional: If time permits, try **combining** two or more techniques (e.g., role-based + few-shot, or chain-of-thought + role-based).

Record Your Observations

Technique	Prompt (brief)	What did the AI produce?	Did it meet your goal?
Zero-shot			
Few-shot			
Role-based			
Chain-of-thought			
Combination (optional)			

 **Add your best prompt(s) to My Prompt Library** – the prompt you liked the most, with a note about which technique(s) you used and why it worked.

Summary

Zero-shot

Give a clear instruction with no examples. Best for simple, well-defined tasks.

Few-shot

Provide examples to show the AI the pattern or format you want it to follow.

Role-based

Assign a persona to shape the AI's tone, vocabulary, and perspective.

Chain-of-thought

Ask the AI to reason step by step for more accurate and transparent answers.

Combining

Mix techniques together for the most powerful and reliable AI outputs.

- 📌 **The Most Important Idea:** Advanced prompting gives you more control. Different techniques work better for different tasks. By trying multiple techniques and comparing, you become a smarter, more effective AI user. Always verify the output and stay in the loop.



Homework: Creative Prompt Lab

You've learned advanced techniques today. Now it's time to **invent your own creative uses** for them. This is your chance to have fun and think outside the box.

Task 1: Pick a Scenario

Choose **one** of these creative scenarios (or invent your own):



Scenario A

A penguin starts a rock band in Antarctica.



Scenario B

Your backpack can talk and it gives terrible advice.



Scenario C

A time-travelling sandwich appears in your lunchbox.



Scenario D

You discover a hidden door in your school library that leads to a world made of candy.

Task 2: Apply Three Techniques

For your chosen scenario, write **three different prompts** using three different techniques from today. Be creative – you don't have to be realistic!

Few-shot

Give 1–2 examples first, then ask for something new.

Example: "Here are two band names: 'The Frozen Chords' and 'Waddle Rock'. Now give me three more band names for a penguin rock band."

Role-based

Make AI act as a funny, weird, or unexpected persona.

Example: "You are a grumpy seal who hates music. Explain why the penguin band's concert is a bad idea."

Chain-of-thought

Ask AI to explain its silly reasoning step by step.

Example: "If the penguin band plays a concert on an iceberg that is melting, think step by step: what could go wrong?"

Next Session Preview

Session 13: Prompting for Real Tasks

Now that you can use advanced techniques, we'll apply them to real-life tasks, for example: creating study guides, writing emails, planning events, and more. You'll also learn about **ethical use** - how to use AI responsibly in school and life.

